

KANGDA WEI

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RESEARCH INTEREST

Natural Language Processing, Large Language Model, LLM Reasoning, LLM Exploratory Thinking, Post-Training, Multi-Modal, Multi-Agent Systems

EDUCATION

Texas A&M University	College Station, TX
<i>Doctor of Philosophy in Computer Science. Advisor: Ruihong Huang.</i>	Aug 2023 - Dec 2027 (Expected)
University of North Carolina at Chapel-Hill	Chapel-Hill, NC
<i>Master of Science in Computer Science. Advisor: Shashank Srivastava.</i>	Aug 2022 - May 2023
University of North Carolina at Chapel-Hill	Chapel-Hill, NC
<i>B.S. in Computer Science, B.S. in Statistics and Operational Research, GPA: 3.812/4.0</i>	Aug 2019 - May 2022

INDUSTRY EXPERIENCE

NLP & Large Language Model Intern	Sunnyvale, CA
<i>Robert Bosch LLC, USA</i>	May 2024 - Aug 2024
<i>Applied Scientist Intern, Mentor: Zhengyu Zhou</i>	

RESEARCH EXPERIENCE

Enhancing Mathematical Reasoning in LLMs via Exploratory Thinking and Diversity-Aware GRPO

- Developed a diversity-promoting reinforcement learning approach that integrates Maximal Marginal Relevance into GRPO to encourage exploration and reduce mode collapse during reasoning optimization.
- Adapted exploratory thinking principles to enhance mathematical reasoning performance across diverse problem-solving styles.
- Achieved an average of 1.5% absolute accuracy improvement across five mathematical reasoning benchmarks (AIME24, AMC23, MATH500, Minerva, and OlympiadBench) compared to both the original GRPO and other diversity-aware GRPO variants.
- *Ongoing:* Conducting ablation and generalization studies; paper under preparation.

Mitigating Bias and Improving Reasoning in LLMs via Exploratory Thinking and RL

- Proposed a novel data generation framework that encourages exploratory thinking in LLMs by prompting them to reason over structurally identical, morally ambiguous story pairs with male and female protagonists.
- Designed a guided judgement reconciliation process to uncover and mitigate biased moral reasoning, producing balanced gender-neutral story-judgement pairs.
- Performed SFT and DPO on LLMs, achieving substantial gender bias reduction on WinoBias and GenMO benchmarks without sacrificing performance on MMLU and TruthfulQA.
- *Ongoing:* Extending framework to other forms of social bias (e.g., race, social status)

Multi-Agent Systems with LLMs/VLMs for Nuclear Safety Anlysis and Video Understanding

- Designed and implemented multi-agent systems that leverage LLMs and VLMs for two domains:
 - *Nuclear Safety:* Developed a BFS-based multi-agent framework for automatic nuclear fault tree generation in collaboration with Idaho National Laboratory (INL). Built a large-scale nuclear safety dataset via OCR and expert validation; benchmarked multiple LLMs/VLMs for fault analysis tasks.
 - *Video Understanding:* Proposed an end-to-end multi-agent multi-modal framework for advanced indexing of lecture/presentation-style videos. Combined VLM-based shot detection with multi-agent collaboration to capture rich cross-modal information for scalable downstream use. Conducted systematic evaluations, including intrinsic human evaluation (Amazon Mechanical Turk) and extrinsic QA benchmarks, on both public and enterprise datasets.

Event Understanding and Coreference in Long Documents

- Introduced *LegalCore*, the first dataset in the legal domain annotated with comprehensive event and event coreference information.
- Benchmarked mainstream LLMs on discourse-level event relation extraction and event coreference tasks, showing that both remain challenging for LLMs.

- Provided strong supervised baselines by fine-tuning smaller language models and conducted extensive empirical analysis, highlighting weaknesses of GPT models in hallucination control, transitivity, and long-range dependency capture.

PUBLICATIONS

- Cheng Zhang, Rajasekhar Kakarla, **Kangda Wei**, Ruihong Huang. ENG-DRB: PDTB-style Discourse Relation Bank on Engineering Tutorial Video Scripts. *In submission*.
- **Kangda Wei**, Hasnat Md Abdullah, Ruihong Huang. Mitigating Gender Bias via Fostering Exploratory Thinking in LLMs. *Findings of the Empirical Methods in Natural Language Processing 2025*.
- Abhilekh Borah, Hasnat Md Abdullah, **Kangda Wei**, Ruihong Huang. CliME: Evaluating Multimodal Climate Discourse on Social Media and the Climate Alignment Quotient (CAQ). *NLP4PosImpact Workshop, Association for Computational Linguistics 2025. Best Paper Award*.
- **Kangda Wei**, Xi Shi, Jonathan Tong, Sai Ramana Reddy, Anandhavelu Natarajan, Rajiv Jain, Aparna Garimella, Ruihong Huang. LegalCore: A Dataset for Event Coreference Resolution in Legal Documents. *Findings of Association for Computational Linguistics 2025*.
- **Kangda Wei**, Zhengyu Zhou, Bingqing Wang, Jun Araki, Lukas Lange, Ruihong Huang, Zhe Feng. Multi-Agent Video Understanding for Advanced Indexing of Lecture Videos. *In submission*.
- Hasnat Md Abdullah, Tian Liu, **Kangda Wei**, Shu Kong, Ruihong Huang. UALBench: The First Comprehensive Unusual Activity Localization Benchmark. *Winter Conference on Applications of Computer Vision (WACV) 2025*.
- **Kangda Wei**, Aayush Guatam, Ruihong Huang. Are LLMs Good Annotators for Discourse-level Event Relation Extraction? *Findings of the Empirical Methods in Natural Language Processing 2024*.
- Jiangshu Du, Yibo Wang, Wenting Zhao, ... **Kangda Wei**, et al. LLMs Assist NLP Researchers: Critique Paper (Meta-)Reviewing. *Proceedings of the Empirical Methods in Natural Language Processing 2024*.
- **Kangda Wei**, Dawn Lawrie, Benjamin Van Durme, Yunmo Chen, Orion Weller. When Do Decompositions Help for Machine Reading? *Proceedings of the Empirical Methods in Natural Language Processing 2023*
- **Kangda Wei**, Sayan Ghosh, Rakesh Menon, and Shashank Srivastava. Leveraging Multiple Teachers for Test-Time Adaptation of Language-Guided Classifiers. *Findings of the Empirical Methods in Natural Language Processing 2023*
- **Kangda Wei**, Sayan Ghosh, and Shashank Srivastava. 2022. Compositional Generalization for Kinship Prediction through Data Augmentation. *Proceedings of the 4th Workshop of Narrative Understanding (WNU2022)*, pages 13–19, Seattle, United States. Association for Computational Linguistics.
- *Songhe Wang, ***Kangda Wei**, Lei Lin, Weizi Li. “Spatial-temporal Analysis of COVID-19’s Impact on Human Mobility: the Case of the United States,” in the *20th and 21st Joint COTA International Conference of Transportation Professionals*. *Co-author: equal contribution

PROFESSIONAL SERVICES

- Reviewer for ACL Rolling Review (ARR) since February 2024 (covering ACL/NAACL/EMNLP/EACL/AACL).
- Reviewer for IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.
- Reviewer for The Computer Journey.

TEACHING SERVICES

Graduate Teaching Assistant:

- COMP211 System Fundamentals, UNC-Chapel Hill, Department of Computer Science, Spring 2023
- COMP431 Internet Services and Protocols, UNC-Chapel Hill, Department of Computer Science, Fall 2022

AWARDS

- Texas A&M University CSE Department Travel Grant, 2023, 2024

SKILLS

- **Programming Languages / Frameworks:** Python, Java, JavaScript, C, HTML, CSS, MATLAB, R, PyTorch, JAX, Hugging Face Transformers, Accelerate, Linux, Parallel Model Training
- **Software & Tools:** PyCharm, VS Code, Spyder, Jupyter Notebook, Weights & Biases,, Docker, Slurm, Git, Tableau, LaTeX, Anaconda, RStudio
- **LLM Training & Post-training:** SFT, RLHF, DPO, GRPO, Reward Modeling, Prompt Engineering, Synthetic Data Generation
- **Cloud & Infrastructure:** AWS, Microsoft Azure, Amazon Mechanical Turk
- **Language:** Chinese (native), English (proficient)